

Luca Lazzaroni

Ricercatore a tempo determinato in Elettronica

DITEN – UNIGE

SEDE Genova, Italia

EMAIL luca.lazzaroni@unige.it

SETTORE RTD-A · IINF-01/A

BIBLIOMETRIA — SCOPUS

68	345	11	21
DOCUMENTI	CITAZIONI	INDICE H	ARTICOLI, 5 ANNI

AGG. 26 OTT 2026

01 Profilo

Ricercatore a tempo determinato (RTD-A) presso il Dipartimento di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni (DITEN), Università di Genova, settore IINF-01/A – Elettronica, dal marzo 2025. Dottore di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni dal 2024.

Attività di ricerca su machine learning embedded ed edge AI: toolchain TinyML, reti neurali quantizzate e binarie su microcontrollori e FPGA, percezione e decisione per la guida automatizzata, deep reinforcement learning, LLM on-device. L'attività si svolge anche nell'ambito di progetti di ricerca europei e nazionali con partner industriali.

`Edge AI` `TinyML` `Deep learning embedded` `Guida automatizzata` `Deep reinforcement learning` `LLM on-device`

02 Contatti e profili

EMAIL luca.lazzaroni@unige.it	UNIGE rubrica.unige.it	ORCID 0000-0001-8092-5473
SCOPUS 57220892898	GOOGLE SCHOLAR w_6a6sYAAAAJ	GITHUB @lucalazzaroni
LINKEDIN luca-lazzaroni		

03 Incarichi e formazione

POSIZIONI

Mar 2025 – oggi

Ricercatore a tempo determinato (RTD-A)

DITEN, Università di Genova

Settore scientifico-disciplinare IINF-01/A — Elettronica.

Ott 2023 — Feb 2025

Assegnista di ricerca

DITEN, Università di Genova

Assegno: “Realizzazione di modelli di Machine Learning per il trattamento di dati provenienti da veicoli autonomi”.

Nov 2020 — Ott 2023

Dottorando di ricerca

DITEN, Università di Genova — XXXVI ciclo

Borsa di dottorato: “Deep learning on the edge”.

FORMAZIONE

2024

Dottorato di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni

Università di Genova — conseguito il 14 marzo 2024

Tesi: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”

2020

Laurea Magistrale in Ingegneria Elettronica (LM-29)

Università di Genova — 24 luglio 2020

Tesi: “Progetto e implementazione di un motore di esecuzione multipiattaforma per applicazioni IoT” · Voto: 110/110

2017

Laurea in Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)

Università di Genova — 27 ottobre 2017

Tesi: “Progetto model-based del controllo di robot Lego in ambiente Simulink”

2014

Diploma di Maturità Scientifica

Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV)

Voto: 100/100

Abilitazione all'esercizio della professione di Ingegnere dell'Informazione — Esame di Stato superato nella seconda sessione 2020.

04 Didattica e incarichi accademici

CORSI

A.A. 2024/25 – 2026/27

Distributed Electronic Systems and Technologies for Environmental Monitoring

Docente titolare · cod. 94631

A.A. 2024/25 – 2026/27

Tiny Machine Learning

Co-titolare · cod. 114752

A.A. 2026/27

Dispositivi e Circuiti Elettronici

Docente titolare · cod. 65997

dall'A.A. 2022/23

Machine Learning for Automated Driving

Cultore della materia · cod. 108716

A.A. 2023/24	**Computer Graphics Arch. — Games and Simulation** Docente a contratto · cod. 172317
A.A. 2022/23	**Autonomous Agents in Games** Supporto alla didattica · cod. 198216
A.A. 2020/21	**Fondamenti di Programmazione di Sistemi Elettronici** Supporto alla didattica · cod. 166061
A.A. 2020/21	**Approccio Makers alla Progettazione Elettronica** Supporto alla didattica · cod. 186985

Tutti i corsi presso l'Università di Genova.

SUPERVISIONE

1 Dottorando — Ali Zare, Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni (XLI ciclo)

12 Tesi di Laurea — Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)

10 Tesi di Laurea Magistrale — Ingegneria Elettronica (LM-29)

7 Tesi di Laurea Magistrale — Ingegneria Tecnologica per la Strategia e la Sicurezza (LM/DS)

2 Tesi di Laurea Magistrale — Informatica (LM-18)

INCARICHI ACCADEMICI

Membro della Commissione Didattica e Ammissione del Corso di Laurea Magistrale in Engineering for Natural Risk Management.

Responsabile della comunicazione per il Corso di Laurea Magistrale in Electronic Engineering.

05 Curatela e servizio

RUOLI EDITORIALI

2026 — **Editor del volume**
APPLEPIES 2026 — Springer LNEE
Atti in corso di pubblicazione.

2025 — **Membro dell'Editorial Board (Early Career)**
Sensors — MDPI
Dal 18 dicembre 2025.

2025 — **Editor del volume**
APPLEPIES 2025 — Springer LNEE
Atti della International Conference on Applications in Electronics Pervading Industry, Environment and Society.

2024

Guest Editor

Energies — MDPI

Special Issue “Tiny Machine Learning for Energy Applications”, aprile 2024 — agosto 2025.

ORGANIZZAZIONE DI CONVEGNI

Program Chair

APPLEPIES 2026

Bologna, Italia, 10–11 settembre 2026.

Program Chair

APPLEPIES 2025

Torino, Italia, 11–12 settembre 2025.

Program Committee

APPLEPIES

Dal 2023.

Program Committee

GALA — Games and Learning Alliance

Dal 2023.

Chairman di sessione

APPLEPIES 2023

1st Short Oral session — Genova, Italia, 28–29 settembre 2023.

ATTIVITÀ DI REVISIONE

RIVISTE

IEEE Access	dal 2023
Journal of Systems Architecture	dal 2023
Engineering Applications of Artificial Intelligence	dal 2024
Sensors	dal 2024
IET Intelligent Transport Systems	dal 2025
IEEE Transactions on Intelligent Vehicles	dal 2026
IEEE Transactions on Circuits and Systems II: Express Briefs	dal 2026
Robotics and Autonomous Systems	dal 2026
Scientific Reports	dal 2026
Journal of Big Data	dal 2026

CONFERENZE

IEEE ICRA — International Conference on Robotics and Automation	dal 2026
IEEE/RSJ IROS — Intelligent Robots and Systems	dal 2025
GALA — Games and Learning Alliance	dal 2023
APPLEPIES	dal 2021

Discover Internet of Things dal 2026

06 Interventi a convegni

13 INTERVENTI

11–12 set 2025	**APPLEPIES 2025** Torino, Italia
11–14 gen 2025	**43rd IEEE ICCE — International Conference on Consumer Electronics** Las Vegas, NV, USA
19–20 set 2024	**APPLEPIES 2024** Torino, Italia
24–26 giu 2024	**55th Annual Meeting of the Italian Society of Electronics (SIE)** Genova, Italia
28–29 set 2023	**APPLEPIES 2023** Genova, Italia
7–9 set 2023	**54th Annual Meeting of the Italian Society of Electronics (SIE)** Noto (RG), Italia
26–27 set 2022	**APPLEPIES 2022** Genova, Italia
7–9 set 2022	**53rd Annual Meeting of the Italian Society of Electronics (SIE)** Pizzo (VV), Italia
21–22 set 2021	**APPLEPIES 2021** Pisa, Italia
1–2 dic 2021	**GALA 2021 — Games and Learning Alliance** La Spezia, Italia (online)
7–9 lug 2021	**52nd Annual Meeting of the Italian Society of Electronics (SIE)** Trieste, Italia (online)
9–10 dic 2020	**GALA 2020 — Games and Learning Alliance** Laval, Francia (online)
19–20 nov 2020	**APPLEPIES 2020** Online

07 Premi e associazioni

2025

Best Presentation of the Session — sessione “Edge Computing”

43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 gennaio 2025

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

ASSOCIAZIONI PROFESSIONALI

IEEE

IEEE Member e IEEE Young Professional.

IEEE Consumer Technology Society

Membro del Technical Committee “Machine Learning, Deep Learning and AI in CE (MDA)”.

SIE — Società Italiana di Elettronica

Socio nell'Unità di Ricerca dell'Università di Genova dal 2021, area di ricerca “Sistemi elettronici ed applicazioni”.

08 Progetti di ricerca

Synergies

Generation of scenarios for development, training, virtual testing and validation of CCAM systems

Europeo — HORIZON-CL5-2023-D6-01-02 · Task Leader, WP8 — Evaluation Framework (T8.1)

Genera scenari di test per addestramento, test e validazione di sistemi CCAM, con focus su traffico urbano ed extraurbano — dove la conoscenza degli scenari rilevanti è molto più scarsa che in autostrada. Coinvolge le principali case automobilistiche europee e molte università del settore.

Hi-Drive

Addressing challenges toward the deployment of higher automation

Europeo — H2020, GA 101006664 · Membro dell'unità di ricerca dell'Università di Genova

Sviluppa e testa su larga scala funzionalità di guida automatizzata avanzata, passando dall'automazione condizionata (SAE L3) a quella alta, in diversi scenari e contesti stradali.

L3Pilot

Piloting Automated Driving on European Roads

Europeo — H2020 · Membro dell'unità di ricerca dell'Università di Genova

Sperimentazione su larga scala di funzioni di guida automatizzata SAE L3 su strade pubbliche europee, con le principali case automobilistiche e fornitori europei.

GEMINI

Green Machine Learning for the IoT

Nazionale — PRIN 2022 · Membro dell'unità di ricerca dell'Università di Genova (DII–UNITN / DITEN–UNIGE)

Un framework per applicazioni TinyML pienamente sostenibili distribuite lungo un continuum edge-to-cloud, pensato per dispositivi embedded senza batteria, inclusi quelli privi di connettività di rete.

ADINA

Assistente Digitale Integrato Automotive

Regionale — POR FESR Piemonte 2014-2020 · Membro dell'unità di ricerca dell'Università di Genova, con Lifetouch

Un assistente virtuale integrato su hardware dedicato a bordo vettura e connesso ai sistemi del veicolo: algoritmi di IA interpretano le richieste del guidatore e agiscono su funzioni veicolo e infotainment.

09 Pubblicazioni

74 VOCI · 68 SU SCOPUS

2026

Cross-platform deployment and characterization of an on-device generative conversational agent
P. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonesi, D.P. Pau
Journal of Systems Architecture vol. 181, art. 103981 Rivista

A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction
L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti
IEEE Signal Processing Letters vol. 33, pp. 136–140 Rivista

Efficient Certified Robustness Assessment for Transformer Models via Verifier-Aware Perturbation-Position Scheduling
V. Soltanmuradov, R. Berta, **L. Lazzaroni**, D.M. Gómez, A. Pighetti, F. Bellotti
Machine Learning and Knowledge Extraction vol. 8, no. 9, art. 290 Rivista

FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction
A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti
IEEE Transactions on Circuits and Systems II: Express Briefs vol. 73, no. 8, pp. 888–892 Rivista

Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms
H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta
IEEE Access vol. 14, pp. 44886–44903 Rivista

StabilEdge, Formal Verification of Edge AI Applications
L. Lazzaroni, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti
IEEE Access vol. 14, pp. 37416–37434 Rivista

Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers
A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 321–327 Conferenza

Preface
M. Ruo Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero
Lecture Notes in Electrical Engineering pp. v–vi Editoriale

Optimizing Binary AutoEncoders on STM32 Microcontrollers
H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 301–307 Conferenza

Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students
L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 242–248 Conferenza

Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform
H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 218–223 Conferenza

Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA
D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 25–31 Conferenza

Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning
O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 212–217 Conferenza

Implementing YoloV8 on Zynq UltraScale+MPSoC FPGA Boards
H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti
Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 289–294 Conferenza

Generation of Synthetic Datasets of Highway Driving Scenarios
M. Cossu, R. Berta, **L. Lazzaroni**, L. Forneris, M. Fresta, A. Pighetti, A. Pilla, J.L. Sauvaget, H. Touil, L. Durville, T. Griffon, G.B. Nejma, F. Hadj Selem, F. Bellotti
IEEE Transactions on Intelligent Transportation Systems Rivista

FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving
A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta
Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 295–301 Conferenza

Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision
A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 85–90 Conferenza

Embedded Deployment of an LLM-Based Voice Assistant
L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 273–278 Conferenza

Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection
V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 286–291 Conferenza

A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors
S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 183–190 `Conferenza`

A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems
E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 249–254 `Conferenza`

A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents
A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 79–84 `Conferenza`

2025

An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection
L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta
IEEE Sensors Journal vol. 25, no. 24, pp. 44859–44874 `Rivista`

TinyML Acceleration with MAX78000
A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta
Proceedings of SIE 2024 Lecture Notes in Electrical Engineering pp. 468–474 `Conferenza`

Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection
L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 156–163 `Conferenza`

Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking
H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 387–394 `Conferenza`

Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking
F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 139–147 `Conferenza`

RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset
R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 395–402 `Conferenza`

RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
IEEE Sensors Letters vol. 9, no. 9, art. 6010204 `Rivista`

Quantization of MobileNetV2 for Resource-Constrained Microcontrollers
L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 497–504 `Conferenza`

Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board
H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 242–249 `Conferenza`

On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection
M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta
IEEE Open Journal of the Industrial Electronics Society vol. 6, pp. 1351–1369 `Rivista`

Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices
A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 323–330 `Conferenza`

Green Machine Learning for the IoT (GEMINI): A Position Paper
L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 339–346 `Conferenza`

Exploring Decision Transformer for Highway Automated Driving
L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 123–130 `Conferenza`

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection
M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta
IEEE Access vol. 13, pp. 26589–26607 `Rivista`

Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection
M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 250–257 `Conferenza`

Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices
A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti
2025 IEEE International Conference on Consumer Electronics (ICCE)
Digest of Technical Papers IEEE International Conference on Consumer Electronics `Conferenza`

Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification
A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 234–241 `Conferenza`

2024

An embedded end-to-end voice assistant
L. Lazzaroni, F. Bellotti, R. Berta
Engineering Applications of Artificial Intelligence vol. 136, art. 108998 `Rivista`

Adversarial Policy Generation in Automated Parking
A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh
ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving `Workshop`

CBin-NN: An Inference Engine for Binarized Neural Networks
F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti
Electronics vol. 13, no. 9, art. 1624 `Rivista`

End-to-End Dataset Collection System for Sport Activities
M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta
Electronics vol. 13, no. 7, art. 1286 `Rivista`

YoloP-Based Pre-processing for Driving Scenario Detection
M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 418–423 `Conferenza`

Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection
M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 444–449 `Conferenza`

Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations
A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 124–129 `Conferenza`

Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity
R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti
Journal of Intelligent and Connected Vehicles vol. 7, no. 3, pp. 229–244 `Rivista`

Developing a TinyML Image Classifier in an Hour
R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti
IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 946–960 `Rivista`

Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving
L. Lazzaroni, A. Pighetti, F. Bellotti, R. Berta
Proceedings of SIE 2023 Lecture Notes in Electrical Engineering pp. 351–356 `Conferenza`

Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge
A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti
IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 781–794 `Rivista`

Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach
L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 352–357 `Conferenza`

Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents
F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 170–180 `Conferenza`

2023

Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles
L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta
International Journal of Serious Games vol. 10, no. 4, pp. 153–170 `Rivista`

Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild
F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion
Applied Sciences vol. 13, no. 11, art. 6432 `Rivista`

Supporting a .csv-based Workflow in MongoDB for Data Analysts
M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta
2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)
IEEE International Symposium on Industrial Electronics `Conferenza`

Investigating High-Level Decision Making for Automated Driving
A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 307–311 `Conferenza`

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations
F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta
IEEE Access vol. 11, pp. 28522–28550 `Rivista`

Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System
M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 9–15 `Conferenza`

Developing a Toolchain for Synthetic Driving Scenario Datasets
M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 222–228 `Conferenza`

Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation
F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta
Proceedings of SIE 2022 Lecture Notes in Electrical Engineering pp. 239–244 `Conferenza`

Deep Reinforcement Learning for Automated Car Parking
L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 125–130 `Conferenza`

2022

Synthetic Driving Scenarios Generator
M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) `Nazionale`

Investigating High-Level Decision Making for Automated Driving
A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) `Nazionale`

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations
F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) `Nazionale`

An Embedded End-to-End Voice Assistant
L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta
53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV) Nazionale

Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications
R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**
Sensors vol. 22, no. 3, art. 713 Rivista

High-Level Decision-Making Non-player Vehicles
A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 223–233 Conferenza

Developing a Synthetic Dataset for Driving Scenarios
J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 310–316 Conferenza

Classifying Simulated Driving Scenarios from Automated Cars
M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 229–235 Conferenza

A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses
F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 43–52 Conferenza

2021

Project and Development of an Italian Offline Embedded Voice Assistant
L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta
52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste Nazionale

Edgine, A Runtime System for IoT Edge Applications
R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**
Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering
pp. 261–266 Conferenza

Adapting Autonomous Agents for Automotive Driving Games
G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio
Games and Learning Alliance
Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics
pp. 101–110 Conferenza

2020

Employing an IoT Framework as a Generic Serious Games Analytics Engine

L. Lazzaroni, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 79–88 Conferenza